

Convex-set extension via ambient interiors

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_of_convex`

theorem

Extends agreement from a convex interior to all source points by continuity.

Statement

Let E, F be real normed spaces and let $f : S \rightarrow T$ be a bijective isometry between subsets. Assume S is convex, its interior in E is nonempty, and

$$x \in \text{int}_E(s) \iff f(x) \in \text{int}_F(t) \quad (x \in S).$$

Then there is a unique surjective real affine isometry $A : E \rightarrow F$ such that $A(x) = f(x)$ for every $x \in S$.

Assumptions

The stated interiors are ambient interiors, not relative interiors in affine spans. Convexity is assumed only for S . Both directions of the displayed interior-membership equivalence are hypotheses for the given f .

Conclusion

The unique ambient affine extension agrees on all of S , including points outside its interior. No separate target convexity or completeness assumption is needed.

Proof route

Restrict the isometry to the two ambient interiors, use the open connected theorem, and extend equality from the dense interior inside the source set.

▼ Proof steps

1. For $x \in \text{int}_E(s)$ the forward implication puts $f(x)$ in $\text{int}_F(t)$. If $y \in \text{int}_F(t)$, let $x = f^{-1}(y) \in S$; the reverse implication puts x in $\text{int}_E(s)$. Thus The isometry restricted to ambient interiors gives a bijective isometry $f : \text{int}_E(s) \rightarrow \text{int}_F(t)$, with the original forward and inverse values.
2. The source interior is convex and nonempty, hence connected, and both interiors are open. These verify every hypothesis of Open connected extension theorem for the restricted isometry. It yields a unique ambient affine isometry A with $A = f$ on $\text{int}_E(s)$.
3. Convexity and nonempty interior give $s \subseteq \overline{\text{int}_E(s)}$. Concretely, choose $c \in \text{int}_E(s)$ and $\rho > 0$ with $B(c, \rho) \subseteq S$. For $x \in S$ and $0 < a \leq 1$, the point $x_a = (1 - a)x + ac$ satisfies $B(x_a, a\rho) \subseteq S$ by convexity, and $x_a \rightarrow x$ as $a \downarrow 0$. Thus the interior points are dense in the subspace S , the density step used in Extension through a dense convex interior.

4. The two functions $A|_S$ and $f : S \rightarrow F$ are continuous. Their equality set is closed in S , because F is a metric space, and contains the dense interior subset. Hence $A(x) = f(x)$ for all $x \in S$. Any other extension restricts to an extension on the source interior; uniqueness from Step 2 makes it equal to A .

Main citations

- Exact theorem and proof
- The isometry restricted to ambient interiors
- Open connected extension theorem

Lean source signature (exact)

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theorem existsUnique_affineExtension_of_convex
  {s : Set E} {t : Set F} (f : s  $\simeq_i$  t)
  (hs : Convex  $\mathbb{R}$  s) (hsint : (interior s).Nonempty)
  (hfi :  $\forall x : s,$ 
     $(x : E) \in \text{interior } s \Leftrightarrow ((f \ x : t) : F) \in \text{interior } t$ ) :
   $\exists! A : E \simeq^{a i}[\mathbb{R}] F,$ 
     $\forall x : s, A \ (x : E) = ((f \ x : t) : F)$ 
```

In the source	Mathematical meaning
$(f : s \simeq_i t)$	The specified bijective isometry between $S \subseteq E$ and $T \subseteq F$, with E, F real normed spaces.
$(hs : \text{Convex } \mathbb{R} \ s)$	The source set is closed under real convex combinations.
$(hsint : (\text{interior } s).\text{Nonempty})$	The ambient interior $\text{int}_E(s)$ contains a point.
$(hfi : \forall x : s,$ $(x : E) \in \text{interior } s \Leftrightarrow ((f \ x : t) : F) \in \text{interior } t$	The next equivalence is required for every point of S , for this same f .
$\exists! A : E \simeq^{a i}[\mathbb{R}] F$	Existence and uniqueness of a surjective real affine isometry of the whole ambient spaces.
$\forall x : s, A \ (x : E) = ((f \ x : t) : F)$	The same extension agrees with f on every point of S , not only its interior.

Exact source and proof.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_of_convex`

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